

LECTURE 30: MATH1061/7861
TOPICS: INTRODUCTION TO COUNTING
DATE: THURSDAY MAY 7, 2026

Learning objectives:

- (1) Solve counting problems, such as counting permutations and selections.

- (c) How many start with a prime number and end with an odd number?

Two cases :

Start with 2 : $\boxed{2} \overbrace{\boxed{} \boxed{} \boxed{} \boxed{} \boxed{}}^{5!} \times \boxed{}^4$

Start with an odd prime :

$\underbrace{\boxed{} \boxed{} \boxed{} \boxed{} \boxed{} \boxed{}}_{5!} \begin{matrix} \boxed{}^3 \\ \boxed{}^3 \end{matrix}$

$$4 \times 5! + 3 \times 3 \times 5! = 13 \times 5!$$

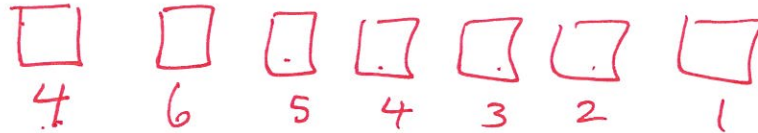
Total number of permutations: $7!$

Q1:

$$= 7 \times 6 \times \dots \times 2 \times 1$$

- (a) How many permutations of $\{1, 2, 3, 4, 5, 6, 7\}$ start with a prime number?

$$4 \times 6!$$



- (b) How many start with a prime number and end with a composite?

4 primes
2 composite

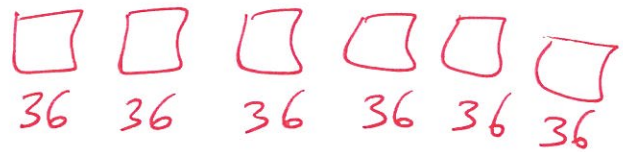


$$\underline{\underline{8 \times 5!}}$$

Q2: In a certain state, number plates consist of six symbols chosen from 26 letters and 10 digits.

- (a) How many distinct number plates are possible if repetition of symbols is allowed?

~~(10 digits)~~



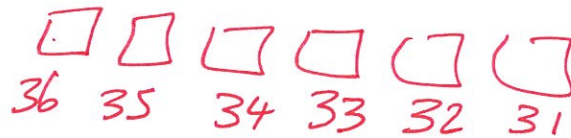
$$36^6 = 2176782336$$

- (b) How many if repetition of symbols is not allowed?

~~(10 digits)~~

ABC 123

ACB 123



$$36 \times 35 \times \dots \times 31 = \frac{36!}{30!}$$

$$1402410240$$

- (b') How many if repetition of symbols is not allowed and order doesn't matter?

$$\frac{36!}{30! \times 6!} = \binom{36}{6}$$

- (c) How many distinct number plates contain at least one repeated symbol?

~~36^6~~

$$36^6 - \frac{36!}{30!} = 774372096$$

- (d) How many number plates start with three digits followed by three letters?

$$10^3 \times 26^3 = 17576000$$

(e) $\rightarrow$ How many if repetition is not allowed?

$$10 \times 9 \times 8 \times 26 \times 25 \times 24 = 11232000$$